

Evaluating Graph-Based and Standard Machine Learning Approaches for Airbnb Pricing Predictions

Project Category: Finance & Commerce

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1 Motivation

This project was inspired by three international first-year master's students who share a passion for traveling. While exchanging travel stories, we noticed that Airbnb consistently emerged as the most popular choice for accommodation, which led us to wonder: *what metrics allow Airbnb to fairly and precisely determine a listing's price?* After some research, we learned that pricing on short-term rental platforms depends not only on listing-level attributes - such as capacity, room type, amenities, and text descriptions - but also on neighborhood spillovers, including proximity to transit, nightlife...

In this project, we frame the task as a regression problem to predict Airbnb listing prices using different machine learning models. Ultimately, we aim to model a city as a graph, where each listing is a node and edges connect geographically close listings. Beyond academic exploration, this project could offer valuable insights for the short-term rental industry by developing a prediction framework that captures the complexities of spatial, textual, and tabular data interactions.

As a reference of one prior research on the subject, we will use the Santorini study that combines GNNs with document embeddings for Airbnb pricing (*Predicting prices of Airbnb listings via Graph Neural Networks and Document Embeddings: The case of the island of Santorini*).

2 Method

We will use the *listings.csv* datasets from **Inside Airbnb** for one city (e.g., San Francisco). Nodes store different features, especially from the Airbnb listings data (e.g., # bedrooms, review scores, room type, neighborhood...) and maybe optional text features from description.

Once the data collected and preprocessed, we will start by using classic algorithms: Linear/Ridge/Lasso, Random Forest, or CatBoost on tabular features, and then a k-nearest neighbors (k-NN) regressor and a geographic k-NN regression. Then we will focus on our own **Graph Neural Network (GNN)** to perform message passing on the spatial graph. Eventually, we might decide to combine the models and work on a "MultiModal-GeoGNN" (tabular + spatial graph + text embeddings).

3 Intended Experiments

Data prep: We will clean and preprocess the Airbnb listings data and construct a feature set from core numeric, categorical and optional text fields.

Evaluation protocol: We will use geographically-aware (and geographically-unaware) train/validation/ test splits to assess generalization to new areas. Performance will be reported with regression metrics and calibration checks. We will keep the test set held out and use it only

once at the very end for final reporting. All model selection and hyperparameter tuning will be done via cross-validation on the training/validation folds.

Expected outcomes: We expect graph-based models to provide gains over tabular baselines.

4 Contributions

We plan to contribute equally to this project, with each team member specializing in one area:

Matthieu: Data collection, preprocessing, transforming the dataset into the desired format, and preparing the train/validation/test split.

Owen: Training and hyperparameter tuning of standard machine learning models (e.g., Linear Regression, KNN, Random Forest, etc.).

Ethan: Training and hyperparameter tuning of Graph Neural Network (GNN) models.

5 References

(Predicting prices of Airbnb listings via Graph Neural Networks and Document Embeddings: The case of the island of Santorini)